CS228 Logic for Computer Science

Exercise Problem Set 4

Part 1

1. Using resolution, show that each of the following sets of clauses is unsatisfiable.
   (a) $H_1 = \{p \vee q,\ \neg p,\ \neg q\}$
   (b) $H_2 = \{p \vee q \vee r,\ \neg p,\ \neg q,\ \neg r\}$
   (c) $H_3 = \{\neg p_1 \vee \neg p_2 \vee \neg p_4,\ \neg p_2 \vee p_4,\ \neg p_1 \vee p_2,\ p_1\}$
   (d) $H_4 = \{a \vee b,\ \neg a \vee b,\ a \vee \neg b,\ \neg a \vee \neg b\}$
2. Using resolution, show each of the following consequences.
   (a) d is a consequence of $G_1 := (\neg a \vee b) \wedge (\neg b \vee c) \wedge (\neg c \vee d) \wedge a$.
   (b) $q \vee r$ is a consequence of $G_2 := (p \vee q) \wedge (\neg p \vee r)$.
   (c) $P_1 \wedge P_2 \wedge P_3$ is a consequence of
   $$F := (\neg P_1 \vee P_2) \wedge (\neg P_2 \vee P_3) \wedge (P_1 \vee \neg P_3) \wedge (P_1 \vee P_2 \vee P_3).$$
3. Consider the CNF formula G with clauses
   $$\begin{aligned} g_1 &= q_1 \vee q_2 \\ g_2 &= \neg q_1 \vee q_3 \vee q_4 \\ g_3 &= q_3 \\ g_4 &= q_4 \vee \neg q_5 \\ g_5 &= q_1 \vee \neg q_3 \vee q_5 \\ g_6 &= \neg q_1 \vee \neg q_4 \end{aligned}$$
   (a) What are the unit clauses in G, if any?
   (b) What are the pure literals in G, if any?
   (c) Let $\pi = \{q_1 \mapsto 0,\ q_2 \mapsto 1\}$ be a partial assignment. What is the status (true / false / unassigned) of each of the following under π?
   - $(q_3 \vee \neg q_1) \wedge (q_1 \vee \neg q_2)$
   - $(q_1 \vee q_2 \vee q_3) \wedge \neg q_1$
   - $q_1 \vee q_4$
   - $\emptyset$ (the empty formula)

   (d) Under the same π, which of the clauses g_2, g_3, g_4, g_6 are unit clauses? Identify the corresponding unit literals.
   (e) Starting from the empty assignment, run the DPLL algorithm (as described in class) on G, applying unit propagation and pure literal elimination whenever possible. Is G satisfiable? If so, give a satisfying assignment.

Part 2

4. A formula is said to be in 2-CNF if it is in CNF and every clause has at most two literals. Show that the satisfiability of any 2-CNF formula can be checked in polynomial time.

5. A clause is called a **Horn clause** if it contains at most one positive literal. A CNF formula, all of whose clauses are Horn clauses, is called a **Horn formula**. For example, $(\neg p \vee \neg q \vee r)$ is a Horn clause (exactly one positive literal, r); so is $(\neg p \vee \neg q)$ (zero positive literals) and the unit clause p (one positive literal). On the other hand, $(p \vee q \vee \neg r)$ is *not* a Horn clause, since it has two positive literals (p and q).

   (a) Which of the following clauses are Horn clauses?

      (i) $\neg p \vee \neg q \vee r$

      (ii) $p \vee q$

      (iii) $\neg p \vee \neg r$

      (iv) $p \vee \neg q \vee r$

   (b) Using resolution, or otherwise, show that the satisfiability of a Horn formula can be decided in polynomial time. What is the time complexity of the algorithm you describe?

   (c) What happens when you run the DPLL algorithm (as described in class) on a Horn formula, always preferring unit propagation over making a decision? How does this relate to your algorithm from part (b)?

6. Using resolution, or otherwise, show that there is a polynomial-time algorithm to decide the satisfiability of those CNF formulas F in which each propositional variable occurs at most twice. Justify your answer.

7. Define *Positive resolution* as a restriction of ordinary resolution as follows: derive a resolvent from clauses C_1 and C_2 only if C_1 is a positive clause, i.e., it consists only of positive literals. Prove or disprove : If F is an unsatisfiable CNF formula then one can derive the empty clause from F using only positive resolution.

8. Let F be an unsatisfiable CNF formula with n propositional variables. Show that there is a resolution proof of $\perp$ from F of size at most $2^{n+1} - 1$ (size = number of clause-occurrences/nodes in the derivation tree).